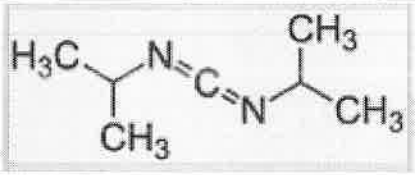
Mylan

Mylan Laboratories Limited
API-R&D – Analytical Development Laboratory

Specifications and Test Procedure	**Doc. No.: STP/RM/ADL/988/01**	Effective Date: 23.12.22
	Supersedes: STP/RM/ADL/988/00	
	Category: Raw material	Page 1 of 3
Structure:		
TITLE: **Diisopropyl carbodiimide**		
Molecular Formula: $C_7H_{14}N_2$		Project: Octreotide Acetate
Molecular Weight: 126.2		Reference: IH

A. Specifications:

S. No	Test Parameter	Specifications
1.0	Description	Clear colorless to light yellow color liquid
2.0	Identification by GC	The retention time of the major peak in the chromatogram of the sample solution corresponds to that in the chromatogram of the standard solution, as obtained in the chromatographic purity.
3.0	Water content (% w/v, by KF)	Not more than 0.5
4.0	Chromatographic purity (% area normalization, by GC) a) Purity b) Any other individual impurity c) Total impurities	 Not less than 98.0 Not more than 1.0 Not more than 2.0

Storage: Store in a well-closed container at 2 to 8°C.

Packing: Pack in a lined metal can or poly liner drum or as recommended by the manufacturer.

Revision Record:

Revision Number	Effective Date	Details of revision
00	19.06.12	First issue
01	23.12.22	Format number has been updated as per revised Specification SOP.

	Prepared by	Reviewed by		Approved by
Sign&Date	23/12/22	V-Suresh 23.12.2022	23/12/2022	Sudha Rani T. 23.12.2022
Name	Tarak Nath Burai	V. Suresh Babu	K. Ananda	T. Sudha Rani
Designation&Dept.	Group Leader-ADL	Group Leader-ADL	Senior Director-CRD	Associate Director-II-ADL

B. Test procedure:

1.0 Description:

By visual observation.

2.0 Identification by GC:

The retention time of the major peak in the sample solution, when analyzed as per the procedure given below for the determination of chromatographic purity, corresponds to that of standard solution.

3.0 Water content (% w/v, by KF):

For the determination of water content in sample, place about 40 mL of dehydrated methanol in the titration vessel and titrate with the standardized reagent to get the electrometric end point. Quickly add about 2.0 mL of sample into a freshly neutralized methanol. Perform the test in duplicate by taking fresh methanol.

Calculation:

$$\text{Water content (\% w/v)} = \frac{V \times F \times 100}{W \times 1000}$$

Where,
V= Volume of KF reagent consumed (mL).
F = Water equivalence factor (mg/mL).
W = Volume of sample (mL).

4.0 Chromatographic purity (% area normalization, by GC):

Instrumentation:

Gas chromatograph: Agilent Technologies 6890N Network GC system equipped with flame ionization detector.
Data handling system: Empower software or its equivalent.

Column:

DB-5 column 30 meters length, 0.53 mm internal diameter, packed with 5 µm film thickness capillary column.

GC Parameters:

Initial oven temperature : 80°C
Initial time : 5 minutes
Rate (1) : 10 °C/minute
Oven temperature (2) : 140°C

Time (2)	: 12 minutes
Rate (2)	: 20°C/minute
Final oven temperature	: 240 °C
Final Time	: 12 minutes
Injector temperature	: 200 °C
Detector temperature	: 260 °C
Carrier gas, Nitrogen flow	: 5.0 mL/minute
Hydrogen flow	: 30 mL/minute
Zero Air flow	: 300 mL/minute
Makeup gas, Nitrogen flow	: 5.0 mL/minute
Split ratio	: 20:1
Injection volume	: 0.2 µL
Run time	: 40 minutes

Standard solution:
Diisopropyl carbodiimide standard as such.

Sample solution:
Sample as such.

Evaluation of System suitability:
Inject standard solution into the chromatograph and record the chromatogram. The system is suitable for analysis, ***if and only if***,
The tailing factor for Diisopropyl carbodiimide peak should be **not more than 1.5**.

Procedure:
Inject separately neat standard and sample solutions into the chromatograph and record the chromatograms. Examine chromatogram of the sample and report the result by percentage area normalization method. The retention time of Diisopropyl carbodiimide is about 11 minutes (only for information).

Calculation for purity: % area normalization.

Calculation for total impurities: 100 - % of purity.

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